Technical Reproducibility Validation Report

mahi-mahi

Version 861ea09
Autosomal STR (ONT)

Verification date	2026-11-01
Repository commit	861ea09ab44e88424c45b04d8018f3948a8fefbb
Result	**Runs as documented**
Reached	Runs + Expected IO
Permalink	https://strhub.app/verified/mahi-mahi-ont
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**Runs as documented**
REACHED	Runs + Expected IO, 4/5 gates passed
WHY	The tool installed and its run produced its documented output.
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. As it is in the repository: Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Runs	Executes end-to-end without crashing	PASS
Expected IO	Produces a non-empty file in the declared format	PASS
Plausible Output	Output structure matches expected genotype-bearing format	—
Execution log	mahi-mahi-ont.log-external.txt	

Stopping at a step is not a finding that the software is faulty. It records how far this particular attempt got. Software can stop early because a dependency it names is no longer available, because it expects input arranged differently from the reference sample, or because the automated environment cannot supply something it needs, such as commercially licensed components. The messages on screen when it stopped are reported below.

3. What the Author Documents as a Known Issue

Quoted from the repository's README at the verified commit. STRhub did not establish any of this by running the tool; it is the author's own note about their own software.

Warnings, Precautions, and Limitations (README line 813)

> ## Warnings and precautions - Use the loci file version that matches the reference genome passed to --ref. - Haplotagged input is recommended. If --bypass-haplotag-check is used, close-in-length alleles may be merged and accuracy may be reduced. - If using non-standard --loci, check the BED file carefully before use. - If a TR region contains sequence that is not actually repetitive, motif decomposition of this unexpected sequence may be effectively nonsensical. The reported length in bases is still meaningful. - Previous versions of mahi-mahi have used different parameters, configurations, an…

Warnings and precautions (README line 815)

- Use the loci file version that matches the reference genome passed to --ref. - Haplotagged input is recommended. If --bypass-haplotag-check is used, close-in-length alleles may be merged and accuracy may be reduced. - If using non-standard --loci, check the BED file carefully before use. - If a TR region contains sequence that is not actually repetitive, motif decomposition of this unexpected sequence may be effectively nonsensical. The reported length in bases is still meaningful. - Previous versions of mahi-mahi have used different parameters, configurations, and filenames. Please refer to…

Known limitations (README line 827)

- When the original BAM contains many non-spanning reads, `DP` can be lower than an allele's `AD` or the sum of `AD`. `DP` measures mean alignment depth over the reference repeat interval, whereas `AD` counts reads selected for allele-length estimation, including selected non-spanning reads. This difference does not by itself indicate an invalid call. - Expansions supported mainly by non-spanning reads may be underestimated and sized less precisely than fully spanned alleles. - Apparent interruptions may be sequencing errors. Visual verification using the read visualisation is recommended. - I…

Troubleshooting and Support (README line 891)

For help, contact Oxford Nanopore Technologies through your usual support channel: [Oxford Nanopore customer support](https://nanoporetech.com/support/customer-support). ## Common checks - Confirm that the reference genome matches the loci file. - Confirm that the BAM or CRAM file is readable and indexed. - Confirm that the sample sex argument is correct. - Confirm that the input BAM is haplotagged unless --bypass-haplotag-check is intentionally being used. ## Common problems and messages | Message or issue | Likely cause | Suggested action | |----------|----------|…

4. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · mahi-mahi 861ea09 · commit 861ea09
$ cd '/opt/tool' &&
touch /tmp/.strhub_mark &&
( mahi-mahi \
--bam /data/in/input.bam \
--ref /data/ref/hg38.fa \
--loci hg38 \
--sample SAMPLE \
--sex XX \
--out-dir out \
); rc=$?; find . \
-xdev \
-type f \
-newer /tmp/.strhub_mark \
-size +0 \
! \
-path './.git/*' \
! \
-path './proc/*' \
! \
-path './sys/*' \
! \
-path './dev/*' \
! \
-path './data/*' \
2>/dev/null |
head \
-500 |
while IFS= read \
-r f; \
do mkdir \
-p "/data/out/$(dirname \
"$f")" &&
cp "$f" "/data/out/$f"; done; exit $rc
```

5. Reproducibility Metadata

Tool	mahi-mahi
Version	861ea09
Repository	https://github.com/nanoporetech/mahi-mahi
Commit (pinned)	861ea09ab44e88424c45b04d8018f3948a8fefbb
Container OS	ubuntu-22.04
Marker panel	Autosomal STR (ONT)
Verified on	2026-11-01 14:11:02 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/41314700506

Verified automatically, in a clean environment, on the tool's public source at the pinned commit. This certificate records what happened; it is not an endorsement by the tool's author.

6. Out of Scope

This report does not evaluate any of the following:

- Genotype correctness or accuracy
- Concordance against known truth sets
- Sensitivity, specificity, or stutter performance
- Allele calling accuracy or forensic casework suitability
- Regulatory compliance or ISO accreditation
- Multi-laboratory or multi-dataset reproducibility

7. Output Content Evidence

Output file	out/SAMPLE.vcf
Format	VCF
Sequence records	99
Distinct STR loci	0
Total reads	0
Max depth (single locus)	0

8. Verification Data

The repository ships example data (2 file(s)); this run did not use it. STRhub ran the verification using the public reference dataset listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

Dataset	1000 Genomes ONT, hg38 CODIS slice (R10 SUP)
Source	https://s3.amazonaws.com/1000g-ont/index.html?prefix=PROCESS…
License	Open access (1000 Genomes / HPRC). Research use. STRhub is not the data provider.
Ref. genome	GRCh38 / hg38
Loci tested	—

9. Verification Matrix

PASS	Reference dataset	1000 Genomes ONT, hg38 CODIS slice (R10 SUP)
	README check (5/5)	Advisory: all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

10. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

– Test data: no sample from the repository was used, so a public reference sample stood in.
– A container environment, built from the tool's declared install steps rather than from a recipe the repository ships.

11. Notes from Reading the Repository

Recorded automatically from the tool's public files when this run was configured. Not verified by execution, and not part of the gates above. Useful for what to check by hand.

– Environment: generated by STRhub from the repository's pyproject.toml (pip), at the pinned commit.
– Input: the README documents BAM (line 55); this run used STRhub's ont-bam-hg38 reference data.
– 1 input path(s) in the README command replaced with /data/in/input.bam.
– 1 reference FASTA path(s) replaced with /data/ref/hg38.fa.
– Run command: the README's own command, rewritten to STRhub's mounts; everything it created was captured as output.

12. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

13. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

14. Conclusion

- **Runs end-to-end**

  mahi-mahi 861ea09 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit.

- **Produces output in the declared format**

  The tool wrote a non-empty VCF file. Whether its contents look like genotype data was not established at this level.

- **Verdict: Runs**

  The tool installed and its run produced its documented output.

- **Repository test data not used**

  The repository ships example data (2 file(s)), which this run did not use: the gates above ran on STRhub's reference dataset. STRhub Verified supplied 1000 Genomes ONT, hg38 CODIS slice (R10 SUP) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

- **Reproducibility statement**

  The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.